EPSR-test-ex120 Bragg peaks, Gaussian broadening, Debye-Waller factor (orthorhombic box)

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the Bragg-peak calculation of the `epsr` engine [1,2,3], which lets EPSR be applied to crystalline samples by adding the sharp Bragg scattering that a small periodic box cannot represent.

Code exercised

Fortran (original, fortran_src/):

- `ato_manipulation_rev4.f`: `bragginten` (reciprocal lattice, hkl list, `.bragg`), `crystfrac`
- `epsr_standalone_rev3.f`: `runepsr17` (keywords; `.inten`, Debye-Waller factor, truncation, replacement of the partial S(Q))
- `ato_manipulation_rev4.f`: `atogen`; `epsr_standalone_rev3.f`: `gensq`

Rust (port, rust/epsr-core/src/):

- `bragg.rs`: `Bragg::read_keywords`, `bragginten`, `structure_and_lattice`, `bragg_peaks`
- `structure.rs`: `atogen`, `gensq`; `epsr.rs`: `Epsr::iterate`

Physical system

A molecular crystal of DMSO and water: a cubic 6.84 Å unit cell with 2 DMSO + 4 water molecules (from `mixato`, `randomise` and 200 Monte Carlo cycles of `epsr` with `potfac 0`), replicated with `makelatticeato` 5 × 4 × 4 into an orthorhombic 34.20 × 27.36 × 27.36 Å box (480 molecules, 2560 atoms, 300 K, written as a general non-cubic box). The atom types are those of EPSR-test, so its `.pcof`, weights and three D-DMSO/water neutron data files are used unchanged; for the crystal they are numerical input only, not a physical model of measured data. For a crystal the simulation box holds only a few unit cells, so the Bragg part of the structure factor [4] is computed from the reciprocal lattice: intensities per atom pair and reflection are broadened into peaks, divided by the number of unit cells, damped by a Debye-Waller factor and replace the intermolecular partial S(Q) up to `qbraggmax`. Gaussian peaks of width 0.01 + fwhmq·Q, Debye-Waller parameter 0.1.

Inputs

`scripts/make_case.sh FORTRAN_BUILD_DIR OUT` (with `make_dmso_water_crystal.sh`, `setkw.py`) regenerates all inputs, verified byte-identical. Keywords changed from EPSR-test: `iinit 1`, `ntimes 2`, `qwidthqmax 0.01 3.0`, `mplicities 5 4 4`, `hklqmin 0.0 10.0 0.1` (all reflections, `rminbragg` 10 Å). `commands.txt`: `repeat 2`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
Per the README `bragginten` enumerates 153 reflections at 8 distinct Q values, and both the Q, Qb > 0 and the Q = 0 broadening branches are taken. The `.bragg` and `.inten` files are compared together with every other output; since the Bragg term replaces the partial S(Q) it also feeds the EP refinement, the fits and all accumulated files. Pass 2 re-reads all files written by pass 1.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] G. L. Squires, Introduction to the Theory of Thermal Neutron Scattering, Cambridge University Press (1978).
[5] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.